

AI Image Generation and NotebookLM: Frequently Asked Questions

This document provides comprehensive answers to common questions regarding the creation of images using Artificial Intelligence and the use of NotebookLM as a personal research assistant, based on the provided lecture materials.

AI Image Generation

How does AI actually create an image from a sentence?

AI image generation uses a process called **diffusion**. Instead of drawing a picture from scratch, the AI starts with a canvas of random static (noise). It then "sculpts" the picture out of that noise by removing the static step-by-step until the final image matches the words provided in the prompt.

Which tools are available for generating AI images?

The following toolkit offers various options depending on whether a user wants to sign up for an account or requires specific features like high-quality text or photorealism: | Tool | Access Requirement | Best For... || ----- | ----- | ----- || **Freegen.app** | No account | Instant, unlimited generation. || **Craiyon** | No account | Generating nine options at once (lower quality). || **Google Gemini** | Google login | Photorealism and accurate text. || **Microsoft Copilot** | Microsoft login | Using DALL-E 3 and strong text. || **Ideogram** | Quick account | Words within images, posters and logos. || **Adobe Firefly** | Quick account | Copyright-safe images (trained only on licensed images). || **Perchance** | No account | Quick access without registration. |

What are the essential ingredients of a great image prompt?

A great prompt is detailed and specific. To gain control over the output, a prompt should ideally include these six ingredients:

1. **Subject:** What is being depicted (e.g., a red fox).
2. **Details:** Specific actions or characteristics (e.g., fluffy, curled up asleep).
3. **Style:** The artistic medium (e.g., watercolour painting, photograph, 3D render).
4. **Composition:** How the shot is framed (e.g., close-up, centered).
5. **Lighting/Mood:** The atmosphere (e.g., soft morning light).
6. **Aspect Ratio:** The dimensions of the image (e.g., 16:9 widescreen).

Why is it important to avoid vague prompts?

The guiding principle for AI imagery is "vague in, vague out". A simple prompt like "a dog" gives the AI too much room to guess, often resulting in a generic or undesired image. A detailed prompt—specifying the breed, the setting (e.g., "running on a beach at sunset"), and camera settings (e.g., "shallow depth of field")—ensures the output matches the user's vision.

NotebookLM: The AI Research Assistant

What is NotebookLM and how does it differ from a standard chatbot?

NotebookLM is a free AI tool from Google designed to be a research assistant. Unlike a standard chatbot that answers based on everything it has ever read (which can lead to invented facts), NotebookLM is **grounded** in your specific sources. | **Feature** | Standard Chatbot | NotebookLM || ----- | ----- | ----- || **Source of Truth** | Everything it was trained on | **Only** the sources you provide || **Accuracy** | May confidently invent facts | Fewer made-up facts (hallucinations) || **Verification** | Difficult to check sources | Every answer includes clickable citations || **Privacy** | Often used to train the AI | Sources stay private; not used for training |

What types of sources can be added to a notebook?

NotebookLM supports a wide variety of formats to ensure it can ground its answers in your specific research material:

- **Documents:** PDFs, Google Docs, Google Slides, and Word files.
- **Web Content:** Websites and YouTube video links.
- **Notes:** Pasted text and copied notes.
- **Audio:** Recorded lectures and voice notes.

What are the limits for sources within a notebook?

Users can upload up to **50 sources per notebook**. While each source can be very long (approximately 500,000 words), it is recommended to use **3–10 focused sources** for better results rather than a large volume of random documents.

What if I don't have sources for a topic yet?

NotebookLM includes a **'Discover'** feature. You can describe a topic, and the tool will find relevant articles from across the web for you to add to your notebook with a single click.

How is the NotebookLM interface organised?

The layout is divided into three functional panels:

1. **Sources (Left Panel):** Displays everything you have uploaded. You can tick specific sources to tell the AI which ones to use when answering a question.
2. **Chat (Centre Panel):** Where you ask questions, get cited answers, and save responses as notes.
3. **Studio (Right Panel):** Contains one-click tools to transform your sources into new formats.

What kind of study materials can the 'Studio' generate?

The Studio panel can turn grounded source material into nine different deliverables:

- **Audio Overview:** A realistic AI-generated podcast.
- **Video Overview:** A narrated slide deck.
- **Mind Map:** An interactive map of ideas and connections.
- **Reports:** Polished documents including Study Guides (key concepts + practice questions), Briefing Docs, FAQs, Timelines, and Blog Posts.
- **Flashcards:** Auto-made sets for self-testing.
- **Quizzes:** Tests with adjustable difficulty levels.

- **Infographics:** Visual one-pagers that can match brand colours.
- **Slide Decks:** Presentation slides downloadable as PPTX files.
- **Data Tables:** Pulls scattered facts into a sortable format.

How does the 'Audio Overview' (AI Podcast) work?

This feature generates a realistic podcast where two AI hosts banter and use analogies to explain your material.

- **Interactive Mode:** You can tap "Join" while the podcast plays to interrupt the hosts and ask a question. They will answer and then resume the discussion.
- **Customisation:** You can set the level (beginner vs advanced) or focus the discussion on a specific section, such as "just chapter four".
- **Language Support:** Podcasts can be generated in over **80 languages** .

Can I collaborate with others on a notebook?

Yes. You can share a notebook with others (they must sign in with a Google account), making it a useful tool for group projects and shared revision. Additionally, NotebookLM is available via a web browser and as an app for both iOS and Android.

What are the known limits of NotebookLM?

- **Garbage In, Garbage Out:** The AI only knows what is in your sources. If the provided material is poor or irrelevant, the output will be as well.
- **Sanity Checks:** While grounding reduces errors, it is still necessary to verify critical information.
- **Free Tier Limits:** Some features, such as the Audio Overviews, may have daily limits.